



Accelerating Manufacturing Excellence

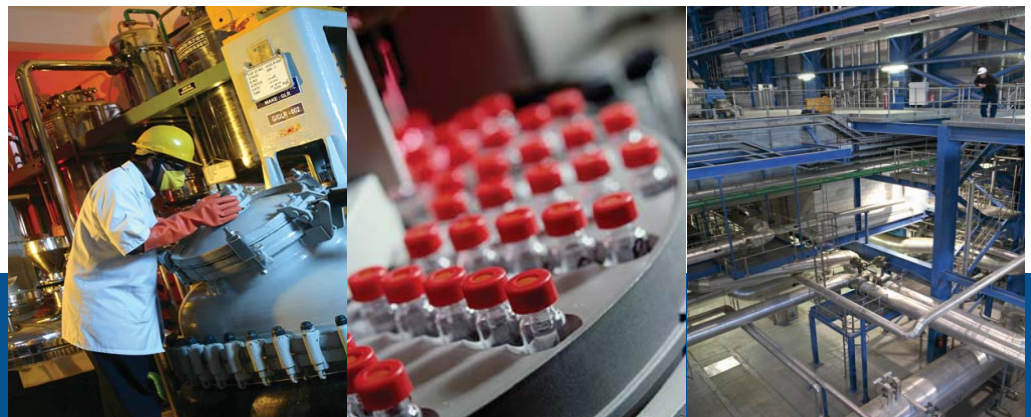


Performix xMES® Suite

Streamlining SAP Manufacturing and Shop Floor Execution

Components:

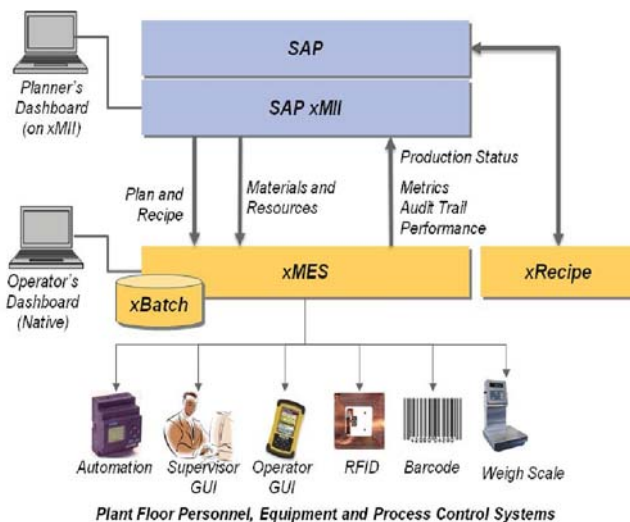
- Performix xMES®
- Performix xRecipe™
- Performix xBatch™



Streamlining SAP Manufacturing and Shop Floor Execution

Performix xMES® Suite

Manufacturing Execution Systems are based on industry standards like the MESA Model, which 'prescribes' an MES to contain certain functionality for the execution of manufacturing processes. The reality today is that many systems used by an organization, such as a SAP/ERP or Automation system, have similar functionality/capabilities for the completion of manufacturing processes. This has led to (a) redundancy and conflict between the Manufacturing IT architecture and the rest of the enterprise, (b) significant amounts of data redundancy, (c) duplication in work, (d) expensive maintenance and support costs. The **Performix xMES®** suite was developed to completely leverage existing functionality residing within the SAP landscape and fill the gaps for MES; thus streamlining and simplifying the shop floor execution of batch manufacturing processes for SAP customers.



The **Performix xMES®** suite consists of several products: **Performix xMES®**, **Performix xRecipe™**, and **Performix xBatch™**. By leveraging the SAP landscape, SAP usability is extended and simplified for shop floor users thus eliminating conflict and duplication in the entire IT architecture and use of SAP Master Data ensuring a "single version of the truth" for the manufacturing enterprise.

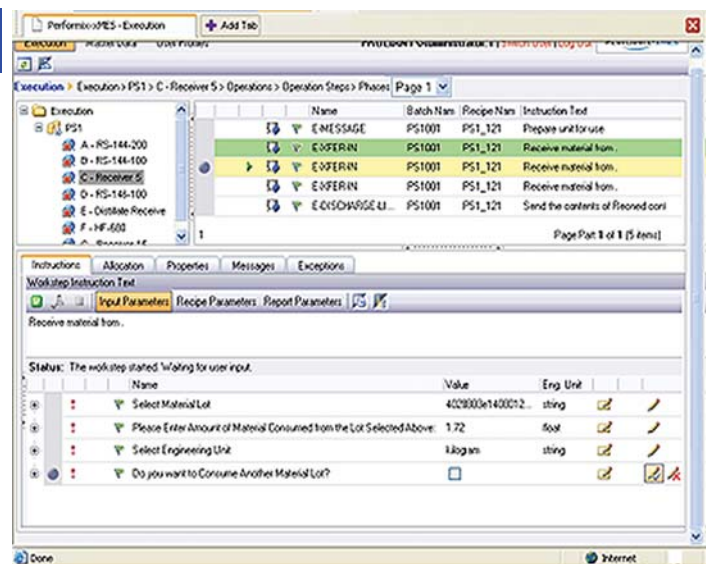
Performix uses industry standards such as B2MML, S95, SAP BAPI's, OPC, and Web Services to integrate with SAP and existing shop floor systems and equipment.

This leads to a significant saving in implementation effort and makes the overall Manufacturing IT architecture much cleaner and less expensive to operate and maintain than traditional, monolithic MES solutions.

Performix xMES® - The Execution Engine

Performix xMES® serves as the "execution engine" for coordinating SAP Batch Manufacturing processes on the shop floor. This is accomplished by downloading the Process Order and Master Data from SAP, executing the Recipe in the process order by providing execution instructions to the Operator, downloading set points to the Automation System, collecting information about the execution in a Historian, and providing up-to-date information on the status and completion of these orders back to SAP.

Performix xMES® leverages **SAP xMII™** transactions conforming to B2MML and S95 standards for bidirectional communication with SAP. It also provides a secure browser based interface to simplify the operator and administrator views of operations and phases which speeds the manufacturing process and enables user authentication for the plant floor execution of a recipe.



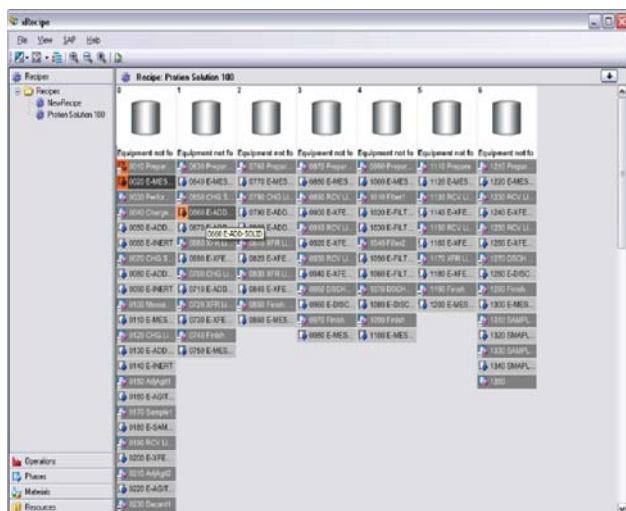
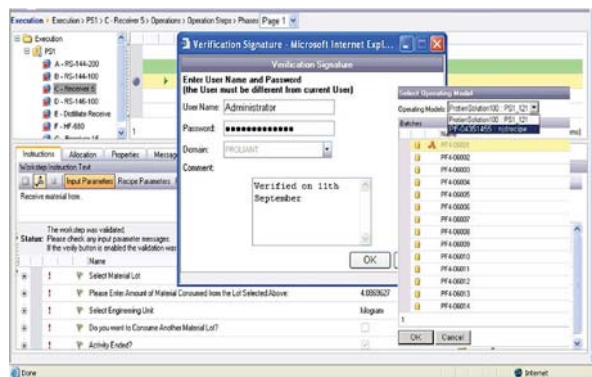
xMES User Interface

The Performix xMES® solution comes with two User Interfaces (UI's); an Administrator UI and an Operator UI.

The **Administrator UI** allows the user to perform system level activities like Process Order, Master Data Download, Recipe Selection, Starting of a Batch, and modifications of a Recipe by adding Ad-Hoc Phases. It is designed to be used from a desktop computer and connects to the xMES server over a Secure Web Service Interface.



The **Operator UI** allows the user to perform all actions required for executing a Recipe at the shop floor, including starting and stopping a phase, entering information associated with phase execution (like material produced or consumed, or any deviations encountered, creating, printing, consuming and destroying Labels). An Operator UI is designed to be used from a Handheld PDA device equipped with a Bar Code and/or RFID reader or Operator panel. (For a list of supported devices, please write us at info@performixinc.com).



Performix xRecipe™ - The Recipe Editor

Editing and Management of Recipes within SAP is a tedious task. Typically, a process engineer must painstakingly create, maintain, and modify recipes in several screens within SAP PP/PI sheets in a process view. Creation or maintenance of recipes in this manner is time consuming and difficult.

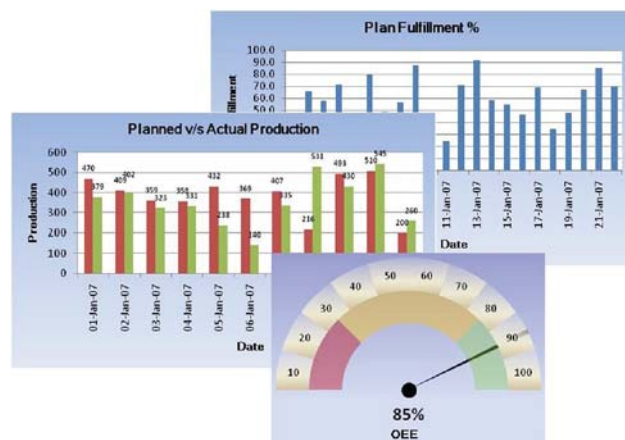
Performix xRecipe™ provides a graphical, drag-and-drop user interface with a visual table, thus providing a full view of each of the workflows and resources required to complete manufacturing execution. An engineer can create a new recipe store it within SAP PP/PI, or, download existing recipes from SAP to edit more quickly and easily.

All data is updated to the SAP PP/PI to allow planners the ability to create the manufacturing schedule. This enhances the usability of SAP PP/PI and simplifies the IT architecture required for the creation and modification of Recipes within SAP.

Performix xBatch™ - the Batch Historian

Important execution information is generated during the execution of a Process/Production Order, such as batch information, material consumed, yield, scrap, resources used, deviations, recipes, etc. All the data captured by the shop floor equipment and devices during the execution of a manufacturing process is stored in **Performix xBatch™** in S88 format. This information is directly integrated into SAP xMII to produce Manufacturing Intelligence Dashboards and manage manufacturing Key Performance Indicators (KPI's) set by management.

Performix xBatch™ also provides a Web Service interface to third party applications for companies that require additional analytics to be performed.





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Performix xMES® Suite Features at a Glance

Performix xMES®

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| <ul style="list-style-type: none">• Download Process Order from SAP (B2MML/S95 compliant xMII Transaction)• Download Master Data from SAP (B2MML/S95 compliant xMII Transaction)• Material Allocation• Select Operating Model/Recipe• Switch Operating Model/Recipe• Modify Operating Model/Recipe• Execute Recipe- Auto/Manual• Log Execution Events in S88 Format in Batch Historian• Edit Master Data• Label Management: Create, Print, Consume and Destroy Labels | <ul style="list-style-type: none">• Label Accounting: Show Label Count in each category.• OPC Integration for connectivity with Automation systems• Synchronization with a DCS System• Formula Execution• Web Service for accessing Batch History• Manufacturing Intelligence/Analytics using xMII• Upload of Production Performance to SAP (B2MML/S95 compliant xMII Transaction)• Material Tracking• Resource Tracking |
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Performix xRecipe™

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| <ul style="list-style-type: none">• Download Master Recipe from SAP (BAPI Interface)• Download Master Data from SAP• Edit SAP Master Recipe using Drag and Drop interface, and Templated Phases & Operations• Preview Recipe in Execution View• Create/Edit Phase and Operation Templates | <ul style="list-style-type: none">• Create/Edit Recipe Parameters and Formulate• Upload the Created/Edited Master Recipe to SAP (BAPI Interface)• Create/Edit Material and Resource Requirements• Create Phase Connections and Phase Networks |
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Performix xBatch™

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| <ul style="list-style-type: none">• Web Service Interface to storing and retrieving execution History• History stored in S88 format | <ul style="list-style-type: none">• Integrated with SAP xMII• Analytics Dashboard generated using xMII |
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Performix, Inc. is based in Houston, Texas with a development center in Pune, India. Performix offers MES solutions for the Pharmaceutical, Process, and Hi Tech manufacturing industry verticals to fill the gaps within enterprise systems and complete the MES solution by leveraging existing ERP and Manufacturing system infrastructure.